

Social Imps Final Paper

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The Right To Repair Fight

Many modern manufacturers employ predatory design practices that intentionally restrict consumers' ability to repair their own products. This includes designing products with proprietary screws, glue-sealed components, or batteries that cannot be easily replaced. These are tactics that effectively force consumers to rely on authorized repair networks or to purchase entirely new devices. As Klosowski explains, this design strategy often makes otherwise functioning electronics obsolete simply because basic repairs are inaccessible or economically unreasonable (Klosowski, 2021). Beyond consumer electronics, such practices are prevalent in agriculture, where companies like John Deere have limited access to diagnostic tools and software necessary for repair. This creates a monopoly on repair services, raising costs and causing harmful delays for farmers who rely on timely equipment maintenance during critical parts of the growing cycle (Gray, 2025).

A 2023 survey found that U.S. farmers lose an average of \$3,348 annually to repair related downtime, with national losses exceeding \$4.2 billion per year (Gray, 2025). This is money that not only farmers have to pay, but everyone who buys their food! These losses are driven not only by higher labor costs at authorized dealers but also by long wait times due to technician shortages. Subscription based repair tools like Deere's Customer Service Advisor charge thousands of dollars annually, while still offering only limited functionality compared to dealer-exclusive software (Gray, 2025). This shift toward software controlled ecosystems not only impedes repair but also positions repairability behind paywalls, further consolidating corporate control. In response, advocacy groups and federal regulators have called for legislation mandating open access to repair tools, parts, and documentation to restore autonomy to consumers and mitigate environmental harm from premature device obsolescence (Klosowski, 2021). In January 2025, the Federal Trade Commission sued

(also in progress) John Deere based on this (Gray, 2025). This is a huge win for not only farmers, but even people who shop at a grocery store. Less crop will be lost, making food cheaper for everyone. This is a clear net positive under utilitarianism, feeding more people for cheaper and wasting less resources at the same time.

From a utilitarian standpoint, which prioritizes the greatest good for the greatest number, the French repairability index offers a significant step toward maximizing societal and environmental well being. By requiring manufacturers to rate and display how easily their products can be repaired based on criteria such as documentation, disassembly, spare parts availability, and cost, the index not only empowers consumers to make more sustainable choices but also encourages producers to design with repair in mind (mdeypere, 2021). This can lead to longer product lifespans, reduced electronic waste, and lower demand for resource intensive manufacturing, all of which benefit society at large. It seems as though the French government is also taking a utilitarianism point of view themselves, often providing detailed formulas for how to calculate the repairability index. For example, the repair.eu website outlines in a lot of detail how to assess the price of spare parts criterion:

"Price of spare parts is a major barrier to repair, but it can fluctuate over time and location. In this case, the geography is restricted to France, and prices declared are the expected prices at the time of compiling the score. The price calculation is based on 3 parameters:

- The price of the most expensive spare part – x
- The average price of the other spare parts – y
- The price of the product – z

The following ratio is computed

$$0.5 * (x + y)/z$$

The points awarded scale linearly from full points if the ratio is less than 10%, to 0 points if the ratio is 30%. Let's assume a product costs 1000 euro, its most expensive spare part costs 300 euro, and the average of the other parts costs 100 euro. The ratio would be

$$0.5 * (300 + 100)/1000 = 0.2 \text{ or } 20\%$$

The score on this criteria would then be half the available points" (mdepypere, 2021).

Despite early skepticism surrounding the French repairability index, particularly concerns about manufacturers rating their own products and the lack of an official verification committee, the initiative has already shown promising impact just one year after its launch. Some critics feared that brands would manipulate scores or conceal important repair data, undermining consumer trust. However, these concerns have not materialized to the extent feared; iFixit, a key participant in the development of the index, reported no blatant falsification of scores, likely due to the reputational and legal risks for noncompliance (Auboy, 2025). Furthermore, the index has positively influenced manufacturer behavior, with companies like Samsung, a major consumer tech company, releasing disassembly manuals to improve their scores. Beyond industry response, public awareness has grown significantly: over half of French consumers are now familiar with the index and consider it when purchasing devices, indicating a shift toward more sustainable consumer ideology (Auboy, 2025). Of course, there are still a lot of criticisms for the index, but it seems like the French government is proactively monitoring the program to make it more reasonable. For instance, on their website it says:

"Question: Isn't reliability an important aspect of a product as well?

Answer: It is! The repairability index will be replaced by a durability index from 1 January 2024. This durability index would combine both repairability and reliability aspects. It will also be an opportunity to tweak the repairability criteria if necessary" (mdepypere, 2021).

In this way, the repairability index has begun fulfilling its utilitarian promise by encouraging better

product design, reducing electronic waste, and empowering consumers all of which contribute to the broader social and environmental good.

A growing body of research shows that expanding the right to repair is not just about consumer rights. It is critical for environmental sustainability too. In the United States alone, nearly 7 million metric tons of e-waste are generated annually (Turiceanu, 2023). This discarded technology contains toxic substances like mercury and cadmium, posing severe health and ecological risks when improperly handled. However, according to Earth.org, a federal right to repair Act could significantly curb this crisis by extending device lifespans and reducing the volume of electronics ending up in landfills (Turiceanu, 2023). Such a shift would create a clear utilitarian benefit: fewer environmental hazards, lower production demands, and better economic outcomes for consumers, who could save up to \$40 billion a year by repairing instead of replacing their devices.

California's 2023 landmark right to repair law, effective from July 2024, marks an important precedent. By mandating open access to repair parts, tools, and manuals for electronics, California paves the way for other states to follow (Turiceanu, 2023). At the same time though, inconsistencies across different state laws highlight the need for national level coherence. Some example include New York's Fair Repair Bill allowing bundled parts sales, or Colorado's focus on farm equipment. Without it, manufacturers can exploit loopholes and delay meaningful reform.

Ultimately, promoting repairability transforms a harmful throwaway culture into a circular economy where devices are valued, maintained, and reused. From a utilitarian perspective, this benefits everyone: consumers save money, small businesses grow through expanded repair services, and the environment suffers less pollution. The right to repair thus emerges not just as a legal battle, but as a crucial step toward a more just and sustainable technological future.

Utilitarianism isn't the only ethical framework that critiques anti-right to repair practices. From a Kantian perspective, which emphasizes duty and respect for persons as ends in themselves, the right to repair also finds strong moral justification. Kantian ethics holds that individuals must be treated not merely as means to corporate profit, but as autonomous agents capable of making informed decisions about their property. When companies restrict access to repair tools or software, they undermine this autonomy by denying consumers the ability to act rationally and independently. Such actions violate Kant's imperative to respect others as rational

beings with their own ends.

Project Contribution and Relevance

Samuel Goldsmith, Brendon Peters, and I created a project based off of the previously mentioned research as well as their own research. Our goal was to increase awareness and radiance for the right to repair, and how important it is even for the average consumer. We made a website (link) that lets an average consumer search for devices and see relevant information about that given device. I created a page where a user can type into an input box and see device names similar to what they typed, as they type. Once a user clicks on one of the devices, it takes them to a new page that contains that particular device's information. I used ifixit's API quite a lot to compile the relevant information on this page, which was inspired by my research. Each page contains an image of the device, a quick summary of the device, and the following information if available:

- The average difficulty of that device's repair guides.
- A download link for that device's French repairability index document.
- An ifixit discussion link for that device.
- A repairability score from ifixit (from 1 to 10).

There were many limitations we met since this was a classroom project. Mainly, we would have liked to analyze a given device's warranty and end user license agreements (EULAs) since they often contain anti-repairability clauses. This proved to be rather difficult however, since manufacturers do not make that information easy to get (at least programmatically).

Future Work

There are several directions for future work. First, expanding the tool to include information on warranty restrictions or EULAs would help users better understand repair limitations. This could involve natural language processing to extract anti-repair clauses. Second, partnering with independent repair networks or aggregating repair pricing data could provide consumers with cost estimates and alternative repair options. Lastly, including links to spare parts on every device's page would be very helpful.

References

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